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Mid-Term Examination, Even Semester 2020-21

B.Tech (CSE)& IBM, II Year, IV Semester

BCSC0005:Computer Organization

Time :2 Hours **Maximum Marks: 30**

Section- A

Note: Attempt All Three Questions. 3 x 2 = 6 Marks

Q1. Convert:

a. $(10110.0101)_2 = (?)_{10}$ b. $(16.5)_{16} = (?)_8$ c. $(26.24)_{10} = (?)_2$

Q2.A digital computer has a common bus system of 64 registers, each with 32 bits. The bus is constructed with a multiplexer.

a. How many multiplexers are needed in the bus?

b. What size of multiplexers are needed?

Q3. Represent the number $(+46.5)_{10}$ in IEEE 754 32-bit floating point format. What would be the bias value for a base-2 exponent in a 6-bit field.

Section- B

*Note: Attempt All Three Questions.*3X3=9 Marks

Q1. Convert the following arithmetic expression into reverse polish notation and show the stack operations for evaluating it.

5*6-[10*(2+6)+8]

Q2. The 8-bit registers AR, BR, CR and DR initially have the following values:AR = 11110010 ; BR = 11111111 ; CR = 10111001 ; DR = 11101010

Determine the 8-bit updated values in the registers after the execution of the following sequence of micro-operations:

AR ← AR+ BR

CR ← CR ∨ DR, BR← BR + 1

AR ← AR ∧ CR

Q3. Perform the operation (-42)-(+13) using signed 2's complement method by using as much bits which are sufficient to represent the numbers in binary. Which gate is used to verify the Overflow condition?

Section – C

Note: Attempt Any Three Questions. 3 x 5 = 15 Marks

Q1. Draw and explain in detail the design of carry look-ahead adder along with its advantage over the binary adder?

Q2. Write a sequence of instructions that will compute the following expression using three-address, two-address and one-address instruction.

$$X = A+B/(C*D*(E-G))$$

Q3. An instruction is stored at location 300 with its address field at location 301. The address field has the value 600. A processor register R1 contains the number 400. With reference to the addressing modes, identify the effective address in each of the following case.

a. Immediate mode

b. Register indirect mode

c. Direct mode

d. Relative mode

e. Index mode with R1 register as index register

Q4. Perform the multiplication of the numbers (+17) and (-10) by using booth's algorithm. Also verify the result.

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Mid-Term Examination, Even Semester 2020-21

B. Tech (CSE-All Specialization), II-Year, IV Semester

Operating System (BCSC0004)

Time: 2 Hours Maximum Marks: 30

Section- A

Note: Attempt All Three Questions. 3 x 2 = 6 Marks

1. What is the role of schedulers? Which scheduler decide the degree of multiprogramming?
2. Explain Non-preemptive and Preemptive CPU Scheduling with example.
3. Which CPU scheduling Algorithm causes Starvation? Write starvation and briefly explain its solution?

Section- B

Note: Attempt All Three Questions 3 x 3 = 09 Marks

1. Define busy waiting condition? How is overcome using semaphore?
2. Distinguish Multiprogramming, Multitasking and Multiprocessing systems.
3. What is the role of kernel in OS? Distinguish clearly the difference between microkernel and monolithic kernel.

Section – C

Note: Attempt Any Three Questions. 3 x 5 = 15 Marks

1. Consider the following processes:

Processes	Burst Time	Arrival Time	Priority
P0	6	0	3
P1	3	2	2
P2	1	3	0
P3	8	5	1

Draw the Gantt Chart for Shortest Job First (SJF- Non-Preemptive) and Preemptive Priority CPU Scheduling. Also Calculate the average response time and average waiting time.

2. What is semaphore. Explain binary and counting semaphore. Write the semaphore based solution for Reader/Writer Classical problem for process synchronization.
3. Explain Operating system as Resource manager! Clearly mention various services provided by Operating system.
4. During the state transition, in which states, the process remains in Secondary memory and main memory? Explain, Seven State process diagram.

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Mid-Term Examination, Even Semester 2020-21

B.Tech (CSE) IInd Year, IVth Semester

BCSC 0015: Applied Data Structures & Algorithms

Time: 2 Hours **Maximum Marks: 30**

Section- A

Note: Attempt All Three Questions. **3 x 2 = 6 Marks**

I. Describe Linked-List Data Structure in comparison with an array. Write the code specification for a Node class of
(a) A Singly Linked-List,
(b) A Doubly Linked-List

II. Study the method signature given below

```
public int countNumberOfNodes(Node node) {
    // write your code here
}
```

Complete the body of the method using a recursive approach, to count the number of nodes in a singly linked-list. Show how to call the method inside the main() method of the Main class.

III. Study the in-fix expression given below
A + B * (C - D) / (P - R)
Write the equivalent reverse-polish notation for the above expression.

Section- B

Note: Attempt All Three Questions. **3 x 3 = 9 Marks**

I. Describe a situation in which you would prefer a linked-list data structure for storing and processing elements instead of a standard array. Define the associated best-case and worst-case time complexities while processing the elements of an array and the nodes of a linked-list, which would be more efficient and why?

II. Explain any two real-life programming situations where stacks are used explicitly. What is the role of a stack in each of the two situations and specify why stack is a better choice for the situation?

III. Explain the measures with which we can observe an algorithm as efficient or inefficient. How are the measures used to deduce if an

algorithm is efficient or not? Are the measures used to deduce the complexity of a Data Structure or an Algorithm, why?

Section C

3 x 5 = 15 Marks

Note: Attempt Any Three Questions(Question 1 is compulsory)

I. Suppose there is an organization, which would like to propose work from home for their employees due to Covid-19 restrictions. An Employee of the organization could possibly choose to work from home, on a first-come, first-served, basis, only the first 'K' employees would be allowed to work from home. Now, observe the signature of the method below and complete the following method.

public Queue allotWorkFromHome(LinkedList employees, int k) {}

The method receives a singly linked list of employee types as 'employees' and the number of employees who would be allowed to switch to work from home as 'k' as parameters. The method should take the first 'k' employees from the linked list, whose and put them into a queue and return the queue of employees. Only those employees should be added to the queue whose 'workingFromHome' field is already set to true in the linked-list.

```
public class Employee {
  private String name;
  private boolean workingFromHome;
}
```

II. Differentiate between a root node, parent node, and a leaf node of a Tree. Differentiate between the algorithm pre-order, post-order and in-order traversal of a Tree.

III. Explain the term "First-In, First-Out" in context to Data Structures. How can you implement a data structure that follows this FIFO approach? Give an example to implement it on an array data structure.

IV. Implement a Stack using an array. The elements in the stack are to be integers. The operations to be supported are PUSH, POP and TRAVERSE. Take into account the conditions of stack overflow and stack underflow.

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Mid-Term Examination, Even Semester 2020-21

B.Tech (CSE/CCV/DA/IoT/CSF/AIML) IInd Year, IVth Semester

BCSC 0006: Data Structures & Algorithms

Time: 2 Hours **Maximum Marks: 30**

Section- A

Note: Attempt All Three Questions. **3 x 2 = 6 Marks**

I. Describe Linked-List Data Structure in comparison with an array. Write the code specification for a Node class of
(a) A Singly Linked-List,
(b) A Doubly Linked-List

II. Study the method signature given below

```
public int countNumberOfNodes(Node node) {
    // write your code here
}
```

Complete the body of the method using a recursive approach, to count the number of nodes in a singly linked-list. Show how to call the method inside the main() method of the Main class.

III. Study the in-fix expression given below
A + B * (C - D) / (P - R)
Write the equivalent reverse-polish notation for the above expression.

Section- B

Note: Attempt All Three Questions. **3 x 3 = 9 Marks**

I. Describe a situation in which you would prefer a linked-list data structure for storing and processing elements instead of a standard array. Define the associated best-case and worst-case time complexities while processing the elements of an array and the nodes of a linked-list, which would be more efficient and why?

II. Explain any two real-life programming situations where stacks are used explicitly. What is the role of a stack in each of the two situations and specify why stack is a better choice for the situation?

III. Explain the measures with which we can observe an algorithm as efficient or inefficient. How are the measures used to deduce if an

algorithm is efficient or not? Are the measures used to deduce the complexity of a Data Structure or an Algorithm, why?

Section C

3 x 5 = 15 Marks

Note: Attempt Any Three Questions(Question 1 is compulsory)

I. Suppose there is an organization, which would like to propose work from home for their employees due to Covid-19 restrictions. An Employee of the organization could possibly choose to work from home, on a first-come, first-served, basis, only the first 'K' employees would be allowed to work from home. Now, observe the signature of the method below and complete the following method.

public Queue allotWorkFromHome(LinkedList employees, int k) {}

The method receives a singly linked list of employee types as 'employees' and the number of employees who would be allowed to switch to work from home as 'k' as parameters. The method should take the first 'k' employees from the linked list, whose and put them into a queue and return the queue of employees. Only those employees should be added to the queue whose 'workingFromHome' field is already set to true in the linked-list.

```
public class Employee {
  private String name;
  private boolean workingFromHome;
}
```

II. Write in detail, the algorithm to delete an element whose index is known in a
(a) Singly Linked-List (b) Double Linked-List

III. Explain the term "First-In, First-Out" in context to Data Structures. How can you implement a data structure that follows this FIFO approach? Give an example to implement it on an array data structure.

IV. Implement a Stack using an array. The elements in the stack are to be integers. The operations to be supported are PUSH, POP and TRAVERSE. Take into account the conditions of stack overflow and stack underflow.

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Mid-Term Examination, Even Semester 2020-21

Program: -B.Tech. (CSE), Year: -II, Semester: -IV

Subject Code: BCHS-0201 Subject Name: -Environmental Studies

Time: 2 Hours **Maximum Marks: 30**

Section- A

Note: Attempt All Three Questions. $3 \times 2 = 6$ Marks

I. Define eutrophication and Bio-magnification.

II. Discuss Environmental Consequences of overexploitation of mineral resources.

III. List the types of natural resources in India.

Section- B

Attempt All Three Questions. $3 \times 3 = 9$ Marks

I. A scientific study reports, that people in some selected villages of Agra are observed with skeletal and bone deformities. Point out your comprehensive view on the problem that the villagers are suffering with its crippling effects

II. Discuss ecological succession with the help of an example of any terrestrial ecosystem.

III. Solar energy is emerging as a viable and reliable source of green energy. Clarify this statement with your views, underlining its importance and limitations

Section – C

Note: Attempt Any Three Questions. $3 \times 5 = 15$ Marks

I. Discuss the vertical structure of atmosphere with the help of a labelled diagram, stating all important activities taking place in different temperature zone regions.

II. What is deforestation? Enumerate and discuss the its various cause and consequences.

III. a.) Discuss different ways for representing the feeding relationship among different level of organisms in an ecosystem. [4]

b.) Explain why the flow of Energy at different levels in an ecosystem is always uni-directional and non-cyclic. [1]

IV. Discuss effects of agriculture and industries on environment.

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Mid-Term Examination, Even Semester 2020-21

B. Tech (CSE, ME), Second Year, IV Semester

BMAS 0103: Engineering Mathematics III

Time : 2 Hours **Maximum Marks: 30**

Section- A

Note: Attempt All Three Questions. 3 x 2 = 6 Marks

Q. 1. Describe Dirichlet's conditions for Fourier series.

Q. 2. Find whether the following partial differential equation is elliptic, parabolic and hyperbolic ?

$$\frac{\partial^2 u}{\partial x^2} + 2x\frac{\partial^2 u}{\partial x \partial y} + (1 - y^2)\frac{\partial^2 u}{\partial y^2} = 0.$$

Q. 3. Find the Laplace transform of

$$e^{-3t}(cos\ 4t + 3\sin 4t).$$

Section- B

Note: Attempt All Three Questions. 3 x 3 = 9 Marks

Q. 1. Find the Laplace transform of

$$\frac{e^{-t}sint}{t}.$$

Q. 2. Expand $f(x) = \pi x - x^2$ in the half-range sine series in the interval $[0, \pi]$.

Q. 3. Find the Laplace transform of $\frac{e^{-at}-e^{-bt}}{t}$.

Section – C

Note: Attempt Any Three Questions. 3 x 5 = 15 Marks

Q. 1. Obtain the Fourier series of function $f(x) = \frac{1}{4}(\pi - x)^2$ in the interval $[0, 2\pi]$. Hence deduce the following relation

$$\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots = \frac{\pi^2}{12}.$$

Q. 2. A string is stretched and fastened to two points l apart. Motion is started by displacing the string in the form $y = A \sin\frac{\pi x}{l}$ from which it is released at time $t = 0$. Show that displacement of any point at a distance x from one end at time t is given by

$$y(x,t) = A \sin\frac{\pi x}{l} \cos\frac{\pi ct}{l}.$$

Q. 3. Describe the solution of heat equation

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$

under the initial condition $u(x,0) = f(x)$ and boundary condtions

$$u(0,t) = 0, u(l,t) = 0.$$

Q. 4. Evaluate

$$\int_0^\infty t^3 e^{-t} \sin t \, dt.$$

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Mid-Term Examination, Even Semester 2020-21
B.Tech CSE/DA/CCV/IOT/AIML/CSF/CSED
Year-II, Semester: IV
BCSC 0010: DISCRETE MATHEMATICS

Time : 2 Hours **Maximum Marks: 30**

Section- A

Note: Attempt All Three Questions. 3 x 2 = 6 Marks

Q1. Find the minimum number of students required in a class to be sure that atleast 8 of them will receive the same grade, if there are 4 possible grades A, B, C and D.

Q2. Use mathematical induction to prove $n^3 + 2n$ is divisible by 3.

Q3. Prove that: (B-A) U (C-A) = (BUC) - A

Section- B

Note: Attempt All Three Questions. 3 x 3 = 9 Marks

Q1. Show that the relation R= {(a, b): a-b = even integer and a, b ε Z}, i.e., aRb $\rightarrow$ a-b = even integer, is an equivalence relation.

Q2. A total of 1232 students have taken a course in Spanish, 879 have taken a course in French, and 114 have taken a course in Russian. Further, 103 have taken courses in both Spanish and French, 23 have taken courses in both Spanish and Russian, and 14 have taken courses in both French and Russian. If 2092 students have taken at least one of Spanish, French, and Russian, how many students have taken a course in all three languages?

Q3. Find the particular solution of the following recurrence relation

$$3a_r = 3a_{r-1} + 10$$

Section – C

Note: Attempt Any Three Questions. 3 x 5 = 15 Marks

Q1. Arun and Tarun appear for an interview for two vacancies. The probability of Arun's selection is 1/3 and that of Tarun's selection is 1/5. Find the probability that

(a) Both of them will be selected.

(b) None of them is selected.

(c) at-least one of them is selected.

(d) Only one of them is selected.

Q2. How many different 8 letter words are possible using the letters of the word SYLLABUS? If a word is chosen at random, find the probability that the word:

(a) Contains two S together.

(b) Begins and ends with L

Q3. Find the generating function for the following sequences:

(a) 2, 2, 2, 2, ...

(b) 0, 0, 3, -3, 3, -3, 3, -3, ...

Q4. Solve the following recurrence relation using generating function:

$$a_n = 8a_{n-1} + 10^{n-1} \quad , a_1 = 9$$

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Mid-Term Examination, Even Semester 2020-21

Programme: B-Tech **Branch:** CS **Year:** II

Computer Networks, BCSC008

Time : 2 Hours **Maximum Marks: 30**

Section- A

Note: Attempt All Three Questions. 3 x 2 = 6 Marks

1. Name the four basic network topologies, and cite advantages of each type.For n devices in a network, what is the number of cable links required for each topology?

2. What is the value of SNR and SNR_{dB} for the noiseless channel? A telephone line uses 400KHz of bandwidth. In this telephone line average signal is 10 watt and the noise is 5 mWatt. What is the maximum data rate supported by this telephone line?

3. Explain in brief the following terms
   (a) Guided Media
   (b) Unguided Media
   (c) Repeater
   (d) Hub

Note: Attempt All Three Questions. 3 x 3 = 9 Marks

1. Name three types of transmission impairment and their causes.
2. What is bit rate and baud rate? Encode the following binary data stream into Manchester and differential Manchester codes 1 1 0 0 1 0 1 0.
3. Compare and contrast among circuit-switched network, packet-switched network and message switched network.

Note: Attempt Any Three Questions. 3 x 5 = 15 Marks

1. What is the total delay (latency) for a frame of size 5 million bits that is being sent on a link with 6 routers each having a queuing time of 5μs and a processing time of 3μs. The length of the link is 5000 Km. The speed of light inside the link is 2×10^8 m/s. The link has a bandwidth of 500 Mbps. Which component of the total delay is dominant? Which one is negligible?
2. How pure ALOHA differ from slotted ALOHA. Discuss the vulnerable time of Pure ALOHA and Slotted ALOHA. One hundred stations on a pure ALOHA network share a l-Mbps channel. If frames are 1000 bits long, find the throughput if each station is sending 10 frames per second.
3. CSMA is based on the principle "sense before transmit", Comment on it. Explain 1 persistent, non-persistent and p persistent method. A network has a data transmission bandwidth of 20×10^6 bits per second. It uses CSMA/CD in the MAC layer. The maximum signal propagation time from one node to another node is 40 microseconds. What is the minimum size of a frame in the network?
4. Explain the OSI reference model in detail.

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Mid-Term Examination, Even Semester 2020-21

Program: B.Tech (CS,CSF,IOT,CCV/DA), Year: II Semester: IV

Subject Code: BCSC-0009 Subject Name: Software Engineering

Time : 2 Hours **Maximum Marks: 30**

Section- A

Note: Attempt All Three Questions. 3 x 2 = 6 Marks

I. Describe the characteristics of software contrasting it with the characteristics of hardware. Illustrate with a diagram that the software does not wear out.

II. Assume that you are the technical manager of a software development organization. A client approached you for a software solution. The problems stated by the client have uncertainties which lead to loss if it not planned and solved. Which software development model you will suggest for this project-justify. Explain the model with its pros and cons and neat sketch.

III. Why do we feel that characteristics of requirements play a very significant role in the selection of a life cycle model? Justify your answer.

Section- B

Note: Attempt All Three Questions. 3 x 3 = 9 Marks

I. Enlist various characteristics of an SRS. Enlist major organization of requirement document recommended by std. IEEE-830-1998.

II. Discuss and Explain Agile Framework with suitable Diagram and advantages and limitations.

III. What do understand by Coupling and Cohesion? Explain three types of coupling and cohesion with example.

Section – C

Note: Attempt Any Three Questions. 3 x 5 = 15 Marks

I. Discuss the elicitation technique which promotes the creative thinking and new ideas. Consider the following activities of Restaurant System:

Customer Order	Serve Product
Collect Payment	Produce Product
Store Product	Order Raw Materials
Pay For Raw Materials	Pay for Labor

Consider the above Scenario and create the following:

a. Use Case Diagram

b. DFD Level-0, level-1

II. Consider a project of GLA University Hostel Management System with following information:

i. External Inputs:
   1. 5 with low complexity
   2. 15 with average complexity

ii. External Outputs:
   1. 17 with high complexity
   2. 2 with low complexity

iii. External Inquiries:
   1. 3 with low complexity
   2. 2 with high complexity

iv. Internal Logical Files:
   1. 15 with low complexity
   2. 4 with average complexity
   3. 5 with high complexity

v. External Interface Files:
   1. 5 with high complexity

In addition to above, system requires moderate master file updating and significant code reusability and other CAF are treated as average. Assume values for different cost drivers are Nominal. Suppose the project is to be developed in C language having size 128 SLOC/FP.

a. Calculate Overall Effort and development time
b. Calculate phase wise effort and development time.

III. What do you understand by structure Design methodology? Apply all four steps of SDM for ATM Machine.

IV. Explain modularization curve with the help of diagram. Compare the Walston-Felix model with the SEL model on a software development expected to involve 6 person-years of effort.

a. Calculate the number of lines of source code that can be produced.
b. Calculate the duration of the development.
c. Calculate the productivity in LOC/PY
d. Calculate the average manning

Lifecycle Phase Values of μ_p

Mode & Code Size	Plan & Requirements	System Design	Detailed Design	Module Code & Test	Integration & Test
Organic Small S=2	0.06	0.16	0.26	0.42	0.16
Organic medium S=32	0.06	0.16	0.24	0.38	0.22
Semidetached medium S=32	0.07	0.17	0.25	0.33	0.25
Semidetached large S=128	0.07	0.17	0.24	0.31	0.28
Embedded large S=128	0.08	0.18	0.25	0.26	0.31
Embedded extra large S=320	0.08	0.18	0.24	0.24	0.34

Lifecycle Phase Values of τ_p

Mode & Code Size	Plan & Requirements	System Design	Detailed Design	Module Code & Test	Integration & Test
Organic Small S=2	0.10	0.19	0.24	0.39	0.18
Organic medium S=32	0.12	0.19	0.21	0.34	0.26
Semidetached medium S=32	0.20	0.26	0.21	0.27	0.26
Semidetached large S=128	0.22	0.27	0.19	0.25	0.29
Embedded large S=128	0.36	0.36	0.18	0.18	0.28
Embedded extra large S=320	0.40	0.38	0.16	0.16	0.30

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Mid - Term Examination, Even Semester 2020-21

B. Tech (CSE), 2nd Year, 4th Semester

BCSC0011: Theory of Automata & Formal Languages

Time: 2 Hours **Maximum Marks: 30**

Section- A

Attempt All Three Questions. [3 x 2 = 6]

1) Design a Finite Automata for L = {w | w ∈ {a, b}* & every *a* is immediately followed by two *b*'s}.

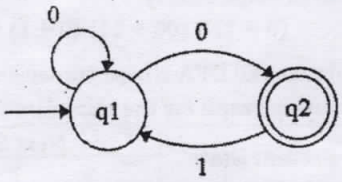
2) Design a Finite Automata that accepts all the decimal strings divisible by 5.

3) Find all strings in L ((a + b)*b(a + ab)*) of length less than four.

Section- B

Attempt All Three Questions. [3 x 3 = 9]

1) Convert the following nondeterministic finite automaton into equivalent deterministic automaton M. Clearly mention all the 5 tuples of M and draw the complete transition graph.



2) Design a PDA for L ={w | $n_a(w) = n_b(w)$}. List all the transitions and also draw the transition graph. [2 + 1]

3) Give Regular Expressions for the following languages.

a) L1 = {0, 11, 101}

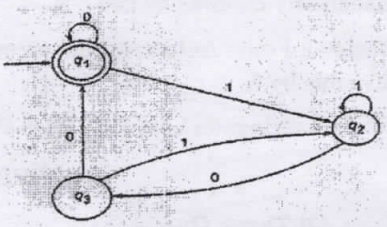
b) L2 = {w | w contains at least three consecutive 1s}

c) L3 = {w | w ∈ {0, 1}* and w begins & ends with the same symbol}

Section – C

Note: Attempt Any Three Questions. [3 x 5 = 15]

1) Apply Arden's theorem to find the Regular Expression corresponding to the following FA



2) Write the statement of pumping lemma for regular set and prove that $L = \{a^k b^k \mid k \geq 0\}$ is not regular. [2+3]

3) Find the DFA for the given regular expression. Show all the required steps clearly.

$$(0 + 1)^* (00 + 11) (0 + 1)^*$$

4) Minimize the DFA whose transition table is given below. Draw the transition graph for the minimized DFA.

Present State	Next State	
	a	b
→q0	q1	q4
q1	q5	q2
q2 (Final state)	q0	q2
q3	q6	q4
q4	q2	q5
q5	q5	q3
q6	q5	q2

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Mid-Term Examination, Even Semester 2020-21

B.Tech, Year: 2nd, Semester: 4th

Introduction to Microprocessors(BCSC 0007)

Time : 2 Hours **Maximum Marks: 30**

Section- A

Note: Attempt All Three Questions. 3 x 2 = 6 Mark

I. What is the use of RESET IN and RESET OUT instruction.
II. Write any four Logic and Bit Manipulation Instructions.
III. Explain Register Indirect Addressing Mode with example.

Section- B

Note: Attempt All Three Questions. 3 x 3 = 9 Marks

I. How AD7-AD0 signal lines are used to transfer both data and low order address using multiplexing? Explain with diagram.
II. Write a program to exchange the contents of memory locations 3000H and 4000H.
III. Explain the following instructions with example:
   1. LXI
   2. SUI
   3. INR

Section – C

Note: Attempt Any Three Questions. 3 x 5 = 15 Marks

I. Draw Timing diagram for the execution of the instruction MVI B,44.(HEX code of instruction MVI B is 06 which is stored at memory location 2001and data 44 is stored at memory location 2002).

II. Explain all the Control and Status Signals of microprocessor 8085.

III. Write a program in microprocessor 8085 assembly language for addition of two 16-bit numbers. Add the 16-bit number in memory locations 4000H and 4001H to the 16-bit number in memory locations 4002H and 4003H. The most significant eight bits of the two numbers to be added are in memory locations 4001H and 4003H. Store the result in memory locations 4004H and 4005H with the most significant byte in memory location 4005H.

IV. What will be the value of all 5 Flag registers and Accumulator after executing each program:

1. MVI A, 86
   MVI B, A7
   ADD B

2. MVI A, 59
   MVI B, 99
   ADD B

3. MVI A, EF
   MVI B, CD
   ADD B

4. MVI A, 79
   MVI B, 5A
   SUB B

5. MVI A, 86
   MVI B, 97
   SUB B

SET-A

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Mid-Term Examination, Even Semester 2020-21

B Tech (All Branches), II Year, IV Semester

Subject Code: English for Professional Purposes (BELH 0004)

Time: 2 Hours **Maximum Marks: 30**

Notes: i. The Question paper has two parts- A & B, with questions on communication skills (13 marks) and verbal ability (17 marks) respectively.
ii. Each question of verbal ability carries 0.33 marks.
iii. Attempt all 3 questions of section A & B, attempt 1 question from section C.

PART A – Communication Skills- 13 marks

Section- A 3 x 1 = 03 Marks

1. Why should one not overuse animation in a slide?
2. What is 'locale'?
3. What is 'Kinesics'?

Section- B 3 x 2 = 06 Marks

4. Write a note on the importance of audience analysis before and during the presentation.
5. What are the strategies of making language polite?
6. 'Technical communication should be accurate.' Explain.

Section – C 1x 4 = 04 Marks

7. Rewrite the following sentences so as to make them suitable for technical communication:
i. The subjects that are considered most important by students are those that have been shown to be useful to them after graduation.
ii. The Backspace key is used to correct errors on the screen.
iii. He has had the physical handicap since he was 5 years old.
iv. Before a surgeon operates, he must know every relevant detail or the patient's history.
8. Write a detailed note on visual aids in a presentation.

Part B- Verbal Ability- 17 mark

Identify the part of speech of the highlighted word (Q 1-15).

1. They wandered **about** sheepskins and goatskins.
A. Noun
B. Adverb
C. Verb
D. Preposition
E. Pronoun

2. Unfortunately, they have not been paying well **of late**.
A. Noun
B. Adverb
C. Verb
D. Adjective
E. Pronoun

3. Ravi didn't tell Shyam anything **except that** Ronit needs a book.
A. Adverb
B. Conjunction
C. Preposition
D. Adjective
E. Verb

4. It's extremely cold **outside**.
A. Noun
B. Adverb
C. Interjection
D. Conjunction
E. None of these

5. An expensive purse was lying **outside** my office room unattended.
A. Noun
B. Adverb
C. Preposition
D. Conjunction
E. Pronoun

6. **Neither** I nor my friend attended the conference last week.
A. Noun
B. Pronoun
C. Conjunction
D. Adverb
E. Preposition

7. One of my relatives **frequents** between Agra and Delhi on almost everyday.
A. Noun
B. Verb
C. Adverb
D. Adjective
E. None of these

8. A torn jeans on sale at Zara went **unnoticed**, entire day.
A. Preposition
B. Adjective
C. Adverb
D. Conjunction
E. Verb
9. **Please,** visit us soon, again!
A. Pronoun
B. Preposition
C. Adverb
D. Conjunction
E. Interjection
10. Your writing is better than **that** of your cousin.
A. Adverb
B. Adjective
C. Pronoun
D. Conjunction
E. None of these
11. I have informed you **why** I cannot take you to the party.
A. Adverb
B. Noun
C. Conjunction
D. Interjection
E. None of these
12. There are more food items **yet** to be tasted.
A. Noun
B. Adverb
C. Pronoun
D. Conjunction
E. None of these
13. There are men of **like** determination and character.
A. Noun
B. Adverb
C. Adjective
D. Preposition
E. None of these
14. On **either** side of the Yamuna Expressway, there are huge signboards.
A. Conjunction
B. Adjective
C. Determiner
D. Adverb
E. None of these
15. The guest is **none** but my old friend.
A. Adverb
B. Adjective
C. Pronoun
D. Noun
E. Determiner

Choose correct answer (Q 16-49)

16. When I went back to the university, I found that a lot of changes_____.
A. were taken place
B. taken place
C. have taken place
D. had taken place
E. None of these
17. She _____ everyday in the morning.
A. is doing yoga
B. was doing yoga
C. did yoga
D. does yoga
E. none of these
18. You _____to smoke inside the Delhi Metro.
A. was not allowed
B. are not allowed
C. had not allowed
D. has not allowed
E. none of these
19. I _____working on this new system software.
A. has been loving
B. was loving
C. am loving
D. love
E. none of these
20. Arvind _____ for his father this morning.
A. has brought a new car
B. have brought a new car
C. had brought a new car
D. brought a new car
E. none of these
21. Nawaz_____ in Vellore since 1990.
A. is teaching
B. was teaching
C. have been teaching
D. has been teaching
E. none of these
22. Before Monsang reached home they _____the police.
A. had already rang up
B. had already rung up
C. have already rang up
D. rung up
E. none of these
23. Dr Puri had been living in Delhi before she _____to Mathura.
A. moved
B. had moved
C. was moved
D. have moved
E. none of these
24. What _____ in Bhubaneshwar?
A. did Kanagarathinam done
B. Kanagarathinam did
C. had Kanagarathinam done
D. did Kanagarathinam do
E. none of these
25. We____ the president on coming Wednesday.
A. have been meeting
B. had been meeting
C. are meeting

D. shall be meeting
E. both c & d

26. Geeta _____ when he was in college.
A. can play the guitar
B. could have play the guitar
C. should have play the guitar
D. could play the guitar
E. None of these

27. She _____ you that she will not attend the meeting
A. should have informed
B. should have inform
C. should had inform
D. should had informed
E. none of these

28. I will discuss this project with my boss when I _____ tomorrow.
A. will meet him
B. would meet him
C. meet him
D. will be meeting him
E. none of these

29. I ____ help you if you require.
A. Will
B. Will be
C. Shall be
D. Would
E. none of these

30. Bala _____ to this conference with his friend next week.
A. will have gone
B. will go
C. should have gone
D. has gone
E. none of these

31. As usual, the_____ of this year's graduates were offered the best jobs.
A. cream of the crop
B. dead wood
C. stool pigeon
D. touch base
E. none of the above

32. If you **play a second fiddle**, you _____
A. accept to be second in importance
B. deny to be second in importance
C. criticize the immediate authority
D. play with juniors
E. none of the above

33. If someone **gets the axe**, he/she _____
A. gets better opportunity
B. will lose his/her job
C. gets the axe to cut trees
D. is given the responsibility of a gardener
E. none of the above

34. The news has already been spread by_____
A. golden handcuffs
B. word of mouth
C. firing lines
D. glass ceiling
E. none of the above

35. If you **show someone ropes**, you _____.
A. explain them how to do a particular job
B. show them the way out of the company
C. explain them how to climb a tree with the help of rope
D. show them a building where pillars are tied with ropes
E. none of the above

36. In truth, I am not from France. I can't speak French.
But if I ____ from France, I _____ French.
A. were, could speak
B. am, can
C. been, could speak
D. were, can speak
E. was, could speak

37. In reality, Afzal has enough money. He won't ask for loan. But if Afzal _____ enough money, he _____ for a loan.
A. had, would ask
B. didn't have, wouldn't ask
C. didn't have, would ask
D. haven't had, would ask
E. None of these

38. Mala is tired. She won't finish her work. But if Mala ___ tired, she ___ her work.
A. weren't, would finish
B. wasn't, would finish
C. weren't, wouldn't finish
D. were, wouldn't finish
E. None of these

39. I didn't go to the office today, so I didn't finish the report.
If I ___ to office today, I ___ my report.
A. hadn't gone, couldn't have finished
B. had gone, could have finished
C. went, finished
D. didn't go, could have finished
E. None of these

40. Yasmeen didn't tell Bhanu about her problem, so he didn't help her. If Yasmeen ___ Bhanu about her problem, he ___ her.
A. had told, could have helped
B. hadn't told, could have helped
C. had told, wouldn't have helped
D. told, would have helped
E. told, could have helped

41. Choose the correct option to fill in the blanks.
If Raamnarayan ___ three hands, he ___ the trays at the cafeteria with two hands and ___ up foods with the other.
A. had, could carry, could pick
B. has, can carry, can pick
C. had has, could have carried, could pick

D. had, could carry, could have picked
E. had had, carried, picked

Choose the grammatically correct sentence (Q 42-47)

42. A. You feel relaxed if you had watched fish swimming in the bank.
B. You will feel relaxed if you watched fish swimming in the bank.
C. You will feel relaxed if you had watched fish swimming in the bank.
D. You feel relaxed if you had been watching fish swimming in the bank.
E. None of these

43. A. If you were in outer space and lit a candle, the flame would perfectly round.
B. If you were in outer space and lit a candle, the flame would be perfectly round.
C. If you were in outer space and lit a candle, the flame had been perfectly round.
D. If you were in outer space and lit a candle, the flame would have been perfectly round.
E. None of these

44. A. If you rub onion juice on your skin, insects stay away.
B. If you rub onion juice on your skin, insects stayed away.
C. If you will rub onion juice on your skin, insects had stayed away.
D. If you would rub onion juice on your skin, insects stay away.
E. None of these

45. A. Did you found the box which I was talking about?
B. Do you find the box which I was talking about?
C. Did you find the box which I was talking about?
D. Did you find the box which I had talking about?
E. None of these

46. Which of the sentences will be appropriate to come after
"Sagar doesn't need a coat. He won't buy one?"
A. If he didn't need a coat, he would buy one.
B. If he didn't need a coat, he wouldn't buy one.
C. If he needed a coat, he wouldn't buy one.
D. If he needed a coat, he would buy one.
E. None of these

47. A. Ramesh had been having too much ice cream before he caught cold.
B. Ramesh have been having too much ice cream before he caught cold.
C. Ramesh have been having too much ice cream before he had caught cold.
D. Ramesh was having too much ice cream before he catches cold.
E. Ramesh had had ice cream before he catches cold.

Identify the erroneous part of the sentence (Q. 48-49)

48. A man who has been (A) / accused of fraud in (B) / an earlier job he will never be (C)/ welcome in any other organization. (D) / No error (E)

49. Identify the erroneous part of the sentence:
I met the gentleman (A)/ today morning (B)/ on my way (C)/to the market. (D). No error (E)

Choose the most logical order of sentences to construct a coherent paragraph (Q 50-51)

50. A. Other than that, the vacation was a success and from then on none of us ever thought there was any place in the world like that lake in Maine.
B. We all got ringworm from some kittens and had to rub Pond's Extract on our arms and legs night and morning.
C. And my father rolled over in a canoe with all his clothes on.
D. One summer, along about 1904, my father rented a camp on a lake in Maine and took us all there for the month of August.
E. I have since become a salt-water man, but sometimes in summer there are days when the restlessness of the tides and the fearful cold of the sea water and the incessant wind which blows across the afternoon and into the evening make me wish for the placidity of a lake in the woods.
A. DBCAE B. ABCDE C. EDCBA D. DBEAC E. BDCAE

51. A. But by around 10,000 years ago, most of North America's animals weighing over 44 kg, also known as megafauna, had disappeared.
B. Before around 10,000 years ago, North America was home to many large and exotic creatures, such as mammoths, gigantic ground-dwelling sloths, larger-than-life beavers, and huge armadillo-like creatures known as glyptodons.
C. The topic has been intensely debated for decades, with most researchers arguing that human overhunting, climate change, or some combination of the two was responsible.
D. Researchers from the Max Planck Extreme Events Research Group in Jena, Germany, wanted to find out what led to these extinctions.
E. With a new statistical approach, the researchers found strong evidence that climate change was the main driver of extinction.
A. BADCE B. DABCE C. BADEC D. EADCB E. BCDAE

SET-B

Printed Pages: 4 **University Roll No.....**

Mid-Term Examination, Even Semester 2020-21
B Tech (All Branches), II Year, IV Semester

Subject Code: English for Professional Purposes (BELH 0004)

Time: 2 Hours **Maximum Marks: 30**

Notes: i. The Question paper has two parts- A & B, with questions on communication skills (13 marks) and verbal ability (17 marks) respectively.
ii. Each question of verbal ability carries 0.33 marks.
iii. Attempt all 3 questions of section A & B, attempt 1 question from section C.

PART A – Communication Skills- 13 marks

Section- A 3 x 1 = 03 Marks

1. What is 'Kinesics'?
2. Why should one not overuse animation in a slide?
3. What is 'locale'?

Section- B 3 x 2 = 06 Marks

4. 'Technical communication should be accurate.' Explain.
5. Write a note on the importance of audience analysis before and during the presentation.
6. What are the strategies of making language polite?

Section – C 1x 4 = 04 Marks

7. Write a detailed note on visual aids in a presentation.
8. Rewrite the following sentences so as to make them suitable for technical communication:

A. The subjects that are considered most important by students are those that have been shown to be useful to them after graduation.
B. The Backspace key is used to correct errors on the screen.
C. He has had the physical handicap since he was 5 years old.
D. Before a surgeon operates, he must know every relevant detail or the patient's history.

PART B- Verbal Ability- 17 marks

Note- Identify the part of speech of the highlighted word **(Q 1-15).**

1. Unfortunately, they have not been paying well **of late**.
A. Noun
B. Adverb
C. Verb
D. Adjective
E. Pronoun

2. They wandered **about** sheepskins and goatskins.
A. Noun
B. Adverb
C. Verb
D. Preposition
E. Pronoun

3. Ravi didn't tell Shyam anything **except that** Ronit needs a book.
A. Adverb
B. Conjunction
C. Preposition
D. Adjective
E. Verb

4. An expensive purse was lying **outside** my office room unattended.
A. Noun
B. Adverb
C. Preposition
D. Conjunction
E. Pronoun

5. it's extremely cold **outside**.
A. Noun
B. Adverb
C. Interjection
D. Conjunction
E. None of these

6. **Neither** I nor my friend attended the conference last week.
A. Noun
B. Pronoun
C. Conjunction
D. Adverb
E. Preposition

7. A torn jeans at Zara went **unnoticed**, all day.
A. Preposition
B. Adjective
C. Adverb
D. Conjunction
E. Verb

8. One of my relatives **frequents** between Agra and Delhi on almost everyday.
A. Noun

B. Verb
C. Adverb
D. Adjective
E. None of these

9. Your writing is better than **that** of your cousin.
A. Adverb
B. Adjective
C. Pronoun
D. Conjunction
E. None of these

10. **Please,** visit us soon, again!
A. Pronoun
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C. Adverb
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11. There are more food items **yet** to be tasted.
A. Noun
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E. None of these

12. I have informed you **why** I cannot take you to the party.
A. Adverb
B. Noun
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13. On **either** side of the Yamuna Expressway, there are huge signboards.
A. Conjunction
B. Adjective
C. Determiner
D. Adverb
E. None of these

14. There are men of **like** determination and character.
A. Noun
B. Adverb
C. Adjective
D. Preposition
E. None of these

15. The guest is **none** but my old friend.
A. Adverb
B. Adjective
C. Pronoun
D. Noun
E. Determiner

Fill in the blanks with the correct option (16-41)

16. She _____ everyday in the morning.
A. is doing yoga
B. was doing yoga
C. did yoga
D. does yoga
E. none of these

17. You _____ to smoke inside the Delhi Metro.
A. was not allowed
B. are not allowed
C. had not allowed
D. has not allowed
E. none of these

18. When I went back to the university, I found that a lot of changes_____.
A. were taken place
B. taken place
C. have taken place
D. had taken place
E. None of these

19. Arvind ____ for his father this morning.
A. has brought a new car
B. have brought a new car
C. had brought a new car
D. brought a new car
E. none of these

20. I _____ this new system software.
A. has been loving
B. was loving
C. am loving
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B. had already rung up
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B. was teaching
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23. Dr Puri had been living in Delhi before she _____ to Mathura.
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B. had moved
C. was moved
D. have moved
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24. We _____ the president on coming Wednesday.
A. have been meeting
B. had been meeting
C. are meeting
D. shall be meeting
E. both c & d

25. What _____ in Bhubaneshwar?
A. did Kanagarathinam done
B. Kanagarathinam did
C. had Kanagarathinam done
D. did Kanagarathinam do
E. none of these

26. She _____ you that she will not attend the meeting.
A. should have informed
B. should have inform
C. should had inform
D. should had informed
E. none of these

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A. can play the guitar
B. could have play the guitar
C. should have play the guitar
D. could play the guitar
E. None of these

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A. Will
B. Will be
C. Shall be
D. Would
E. none of these

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A. will meet him
B. would meet him
C. meet him
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B. dead wood
C. stool pigeon
D. touch base
E. none of the above

31. Bala _____ to this conference with his friend next week.
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B. will go
C. should have gone
D. has gone
E. none of these

32. If you **play a second fiddle,** you _____
A. accept to be second in importance
B. deny to be second in importance
C. criticize the immediate authority
D. play with juniors
E. none of the above

33. If someone **gets the axe**, he/she ___.
A. gets better opportunity
B. will lose his/her job
C. gets the axe to cut trees
D. is given the responsibility of a gardener
E. none of the above

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A. golden handcuffs
B. word of mouth
C. firing lines
D. glass ceiling
E. none of the above

35. If you **show someone ropes**, you ____.
A. explain them how to do a particular job
B. show them the way out of the company
C. explain them how to climb a tree with the help of rope
D. show them a building where pillars are tied with ropes
E. none of the above

36. In reality, Afzal has enough money. He won't ask for loan. But if Afzal ___ enough money, he _____ for a loan.
A. had, would ask
B. didn't have, wouldn't ask
C. didn't have, would ask
D. haven't had, would ask
E. None of these

37. In truth, I am not from France. I can't speak French. But if I _____ from France, I _____ French.
A. were, could speak
B. am, can
C. been, could speak
D. were, can speak
E. was, could speak

38. Mala is tired. She won't finish her work. But if Mala ___ tired, she ___ her work.
A. weren't, would finish
B. wasn't, would finish
C. weren't, wouldn't finish
D. were, wouldn't finish
E. None of these

39. Yasmeen didn't tell Bhanu about her problem, so he didn't help her. If Yasmeen ___ Bhanu about her problem, he ___ her.
A. had told, could have helped
B. hadn't told, could have helped
C. had told, wouldn't have helped
D. told, would have helped
E. told, could have helped

40. I didn't go to the office today, so I didn't finish the report.
If I ___ to office today, I ___ my report.
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B. had gone, could have finished
C. went, finished
D. didn't go, could have finished
E. None of these

41. If Raamnarayan _____ three hands, he _____ the trays at the cafeteria with two hands and _____ up foods with the other.
A. had, could carry, could pick
B. has, can carry, can pick
C. had has, could have carried, could pick
D. had, could carry, could have picked
E. had had, carried, picked

Choose the grammatically correct sentence (Q 42-47)

42. A. If you were in outer space and lit a candle, the flame would perfectly round.
B. If you were in outer space and lit a candle, the flame would be perfectly round.
C. If you were in outer space and lit a candle, the flame had been perfectly round.
D. If you were in outer space and lit a candle, the flame would have been perfectly round.
E. None of these

43. A. You feel relaxed if you had watched fish swimming in the bank.
B. You will feel relaxed if you watched fish swimming in the bank.
C. You will feel relaxed if you had watched fish swimming in the bank.
D. You feel relaxed if you had been watching fish swimming in the bank.
E. None of these

44. A. If you rub onion juice on your skin, insects stay away.
B. If you rub onion juice on your skin, insects stayed away.
C. If you will rub onion juice on your skin, insects had stayed away.
D. If you would rub onion juice on your skin, insects stay away.
E. None of these

45. A. Did you found the box which I was talking about?
B. Do you find the box which I were talking about?
C. Did you find the box which I was talking about?
D. Did you find the box which I had talking about?
E. None of these

46. A. Ram had been having too much ice cream before he caught cold.
B. Ram have been having too much ice cream before he caught cold.
C. Ram have been having too much ice cream before he had caught cold.
D. Ram was having too much ice cream before he catches cold.
E. Ram had had ice cream before he catches cold.

47. Choose the most appropriate sentence to follow, *'Sagar doesn't need a coat. He won't buy one?'*
A. If he didn't need a coat, he would buy one.
B. If he didn't need a coat, he wouldn't buy one.
C. If he needed a coat, he wouldn't buy one.
D. If he needed a coat, he would buy one.
E. None of these

Identify the erroneous part of the sentence (Q 48-49):

48. A man who has been (A) / accused of fraud in (B) / an earlier job he will never be (C)/ welcome in any other organization. (D) / No error (E)

49. I met the gentleman (A)/today morning (B)/ on my way (C)/to the market. (D)/No error (E)

Note - *Choose the most logical order of sentences from the given choices to construct a coherent paragraph (Q 50-51).*

50. A. But by around 10,000 years ago, most of North America's animals weighing over 44 kg, also known as megafauna, had disappeared.
A. Before around 10,000 years ago, North America was home to many large and exotic creatures, such as mammoths, gigantic ground-dwelling sloths, larger-than-life beavers, and huge armadillo-like creatures known as glyptodons.
B. The topic has been intensely debated for decades, with most researchers arguing that human overhunting, climate change, or some combination of the two was responsible.
C. Researchers from the Max Planck Extreme Events Research Group in Jena, Germany, wanted to find out what led to these extinctions.
D. With a new statistical approach, the researchers found strong evidence that climate change was the main driver of extinction.

A. BADCE B. DABCE C. BADEC D. EADCB E. BCDAE

51. A. Other than that, the vacation was a success and from then on none of us ever thought there was any place in the world like that lake in Maine.
B. We all got ringworm from some kittens and had to rub Pond's Extract on our arms and legs night and morning.
C. And my father rolled over in a canoe with all his clothes on.
D. One summer, along about 1904, my father rented a camp on a lake in Maine and took us all there for the month of August.
E. I have since become a salt-water man, but sometimes in summer there are days when the restlessness of the tides and the fearful cold of the sea water and the incessant wind which blows across the afternoon and into the evening make me wish for the placidity of a lake in the woods.

A. DBCAE B. ABCDE C. EDCBA D. DBEAC E. BDCAE

Printed Pages: **University Roll No.....**

Mid-Term Examination, Even Semester 2020-21

Branch: B.Tech. CS/EC, Year: II/III, Semester: IV/VI

BCSE-0903, Data Sciences and Analytics

Time: 2 Hours **Maximum Marks: 30**

Note: Attempt All Three Questions. 60 x 0.5 = 30 Marks

1. To start Python from the command prompt, use the command _____

a) Execute python
b) Go python
c) python
d) run python

2. Which of the following functions is a built-in function in python language?

a) val()
b) print()
c) printls()
d) None of these

3. Study the following function:

>>> round(4.576)

What will be the output of this function?

a) 4
b) 5
c) 576
d) 5

4. Study the following statement:

>>>"a"+"bc"

What will be the output of this statement?

a) a+bc
b) abc
c) a bc
d) a

5. Choose the command for file system from following in Rpi system :

a) ls-
b) lst-s
c) ls
d) l-s

6. Which one of the following is not an HTTP Method:

a) DELETE
b) GET
c) POST
d) UNDO

7. ------- command displays the name of the present working directory:

a) pwds
b) pwd
c) pdw.
d) pw-d

8. You can use ---------to create a new directory:

a) makdir
b) mkdir
c) mkdirs
d) mkdirec

9. To remove directories, use .....

a) rmdr
b) rem-dir
c) remdir
d) rmdir

10. Using .....makes a copy of a file and places it at the specified location:

a) cpi
b) cop
c) cpy
d) cp

11. The .....command moves a file and places it at the specified location

a) move
b) mv
c) mov
d) mve

12. You can use ..... to list the contents of file(s)

a) cat
b) ct
c) cts
d) cnt

13. The ----- command displays the beginning of a file.

a) head
b) hds
c) hde
d) heads

14. ------ denotes the secure shell. Connect to another computer using an encrypted network connection.

a) sshe
b) sshs
c) ssh
d) sesh

15. The ------ command enables you to run a command as a superuser, or another user.

a) suddo
b) su-do
c) sudo
d) sudos

16. --------- command is used to check that whether communication can be made with another host or not.

a) pin-g
b) Ping
c) Pi-ng
d) pings

17. ----------command will tell you the network configuration details such as your IP address.

a) ifconfig
b) ipconfig
c) ip-config
d) if-config

18. ---- command is used to install, upgrade and remove the packages.

a) apt-get
b) aptget
c) aptgets
d) aptgets

19. ------- command will open the configuration settings menu.

a) raspiconfig
b) raspi-config
c) ras-piconfig
d) raspicon-fig

20. -----: Upgrades all of the software packages you have installed.

a) apt-gets upgrade
b) apts-get upgrade

c) aptget upgrade
d) apt-get upgrade

21. ----------Synchronizes the list of packages on your system to the list in the repositories.

a) apt-gets update
b) apts-get update
c) aptget update
d) apt-get update

22. ---------- To shut down immediately.

a) poweroff
b) power-off
c) pwroff
d) shdwn

23. ---------: Prints the current date.

a) date
b) dt
c) date-
d) date-h

24. what is the processing speed of the Raspberry pi 3 model.

a) 700 mhz
b) 850 mhz
c) 1.2 ghz
d) 1.4 ghz

25. python prompt is represented as

a) >
b) >>
c) >>>
d) $

26. _____ is the example of Active RFID tag.

a) Metro token
b) Fast-tag
c) MFRC522-Tag
d) MFRC599-Tag

27. REST stands in RESTful HTTP method:

a) Representational State Transfer
b) Representational Status Transfer
c) Representational State Transmission
d) None of the above

28. ------Successful server response of HTTP protocol :

a) 301
b) 400
c) 399
d) 200

29. Which out of below option is not a property parameters of Thing:

a) Logged
b) Persistent
c) Read Only
d) TimeStamp

30. Name the service used to access historical data from database.

a) QueryPropertyHistory
b) QueryRequestHistory
c) QuerySetProperty
d) None of the above

31. Find the perfect match:

GPIO.setmode(GPIO.BCM)

Pin = 4

Here Pin = 4 will represent:

a) GPIO PIN
b) Physical Pin

32. How many analog pins are available in RPi.

a) 1
b) 2
c) 3
d) None of the above

33. Select the property used to fetch the data to Mashup widgets.

a) GetProperties
b) SetProperties
c) GetsProperties
d) Get-Properties

34. Thingworx is the product of ------- company.

a) PTTC
b) PTC
c) DASSAULT
d) RENOULT

35. 401 HTTP response stands for:

a) Unauthorized access
b) Page not found
c) Method not allowed
d) None of above

36. Select the right function to read humidity and Temperature using DHT11 in Rpi.

a) read_dht11
b) read_retry
c) read_try
d) None of above

37. If input to the Voltage Sensor is 22 V , what will be the appeared potential at the output terminal.

a) 4.9v
b) 4.4v
c) 4.3v
d) 4.88v

38. Voltage Sensitivity of LM35 is

a) 100mV/ °C
b) 10mV/°C
c) 12mv/°C
d) 160 mV/°C

39. Type of communication protocol supported by RFID

a) UART
b) UART and I2C
c) UART, I2, and SPI
d) None of the above

40. Select the function used to return the Timestamp of Rpi

a) time.time()
b) time_time()
c) time as time()
d) time.tm()

41. what does sudo reboot does

a) Shut down
b) File open
c) Restart
d) None of them

42. Which pins are used for SPI connection with RPi interfacing with RFID

a) RX and TX
b) MISO and MOSI
c) SS, SCK, MISO and MOSI
d) SDA and SCL

43. Is Python case sensitive when dealing with identifiers?

a) yes
b) no

44. Which of the following data types is not supported in python?

a) String
b) Numbers
c) Slice
d) List

45. Select the proper time delay given in the following line of code:

time.sleep(0.1)

a) 100 mS
b) 10 mS
c) 1 mS
d) 0.1 mS

46. Raspberry Pi3 is consists of an inbuilt Bluetooth system.

a) True
b) False

47. How many GPIO pins RPi 3B have?

a) 24
b) 18
c) 28
d) 22

48. Name the communication interface used in Raspberry Pi 3?

a) UART
b) SPI
c) I2C
d) All of them

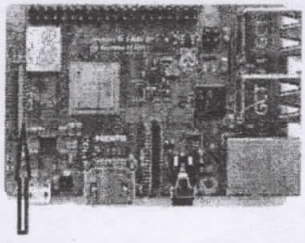
49. What is the Processor operating voltage of RPi3?

a) 3.3V
b) 4.1V
c) 5.2V
d) 3.2V

50. What is the raw voltage input given to RPi3?

a) 5V,2A
b) 5V,3A
c) 2V,5A
d) 4V,4A

51. The Arrow pointing Part of the RPi stands for?



a) Ethernet controller
b) DSI Display port
c) Video Port
d) Dual band WIFI

52. DNS stands for .....

a) determine name system
b) domain name system
c) device name system
d) development name system

53. RISC stands for .....

a) reduced instruction set architecture
b) rare information set architecture
c) reduce information set architecture
d) rare instruction set architecture

54. API stands for ..... .

a) application programming interface
b) Android programming interface
c) Arduino protocol information
d) application protocol interface

55. Why requests library is used? Select that applies:

a) Standard for making HTTP requests in Python
b) Provides a clear and concise framework to work with HTTP requests.
c) Abstracts the complicated parts of HTTP protocol.
d) None of the above

56. 404 response code for a HTTP request stands for:

a) Page not found
b) Unauthorized Access
c) Method not allowed
d) None of the above.

57. "requests.put() function is used to:

a) Receive a data
b) Send a data
c) Both of the above

d) None of the above.

58. Select the temperature range measured by LM35:

a) -20 °C to 100 °C
b) -55 °C to 150 °C
c) -10 °C to 150 °C
d) -20 °C to 150 °C

59. Select the voltage range that can be measured by Voltage sensor:

a) 0 V – 10 V
b) 0 V – 25 V
c) 0 V – 12 V
d) 0 V – 9 V

60. Select the correct formula for voltage divider Circuit:

a) Vout = Vin (R2 / (R1 + R2))
b) Vout = Vin ((R1 + R2) / R2)
c) Vout = Vin (R1 / R2)
d) None of the above

Paper Code-BTDH 0302	SET-C

Soft Skills

Mid Term, Even Sem Examination (2020-21)

Course- B.Tech **Year- II**

Branch- (All)

Time : 120 Min. **M.M-40**

Instructions :

1. **Attempt all questions.**
2. **There will be no negative marking. Also, no marks will be awarded if more than one answer is marked.**
3. **The OMR form is printed on a special paper and the information given by the candidates will be scanned. Thus, the candidates are required to fill the forms carefully. Mutilated and damaged forms will be rejected by the scanner.**
4. **The candidates should ensure that the form is not folded, crumpled or mutilated in any manner. The candidates should not mark anything anywhere except wherever required. The black marks on the margin of the OMR form are important. The candidates should not tamper it with marker.**
5. **Only black& blue pen should be used to fill forms.**
6. **Each branch is identified with a specific subject code which should be written in subject column of OMR sheet**
7. **Write your University roll number in the roll no column of OMR sheet.**

Name of Student:-.....

Branch:-.....

Section-.....

Uni. Roll No:.....

All The Best

(Aptitude)-No of question-80,

Q1. If South-East becomes North, North-East becomes West and so on. What will West become?
A) North-East B) North-West C) South-East D) South-West
E) None of these

Q2. One morning after sunrise, Suresh was standing facing a pole. The shadow of the pole fell exactly to his right. To which direction was he facing?
A) East B) West C) South D) Data Inadequate
E) None of these

Q3. Rahul put his timepiece on the table in such a way that at 6 P.M. hour hand points to North. In which direction the hour hand will point at 9:15 P.M.?
A) South-East B) South C) North-West D) West E) None of these

Q4. Rasik walked 20 m towards north. Then he turned right and walks 30 m. Then he turns right and walks 35 m. Then he turns left and walks 15 m. Finally he turns left and walks 15 m. In which direction and how many meters is he from the starting position?
A) 15 m West B) 30 m East C) 30 m West D) 45 m East
E) None of these

Q5. A man walks 5 km toward south and then turns to the right. After walking 3 km he turns to the left and walks 4 km. And then he goes back 10 km straight. Now in which direction is he from the starting place?
A) South-East B) North-West C) South D) West E) None of these

Q6. K is 40 m South-West of L. If M is 40 m South-East of L, then M is in which direction of K?
A) East B) West C) North-East D) South E) None of these

Q7. A man is facing north-west. He turns 90° in the clockwise direction, then 180° in the anticlockwise direction and then another 90° in the same direction. Which direction is he facing now?
A) South-East 0B) West C) North-East D) South E) None of these

Q8. Golu started from his house towards North. After covering a distance of 8 km he turned towards left and covered a distance of 6 km. What is his distance now from his house?
A) 10 KM B) 16 KM C) 14 KM D) 2 KM E) None of these

Q9. Hemant in order to go to university started from his house in the east and came to a crossing. The road to the left ends in a theatre, straight ahead is the hospital. In which direction is the university?
A) North B) South C) East D) Can't be determined
E) None of these

Q10. Reena walked from A to B in the East 10 feet. Then she turned to the right and walked 3 feet. Again she turned to the right and walked 14 feet. How far is she from A?
A) 4 feet B) 5 feet C) 24 feet D) 27 feet E) None of these

Q11. After walking 6 km, I turned to the right and then walked 2 km. After then I turned to the left and walked 10 km. In the end, I was moving towards the North. From which direction did I start my journey?
A) North B) South C) East D) West E) None of these

Q12. Ravi left home and cycled 10 km towards South, then turned right and cycled 5 km and then again turned right and cycled 10 km. After this he turned left and cycled 10 km. How many kilometers will he have to cycle to reach his home straight?
A) 10 km B) 15 km C) 20 km D) 25 km E) None of these

Q13. A child went 90 m in the East to look for his father, then he turned right and went 20 m. After this he turned right and after going 30 m he reached to his uncle's house. His father was not there. From there he went 100 m to his north and met his father. How far did he meet his father from the starting point?
A) 80m B) 100m C) 140m D) 260m E) None of these

Q14. Shyam walks 5 km towards East and then turns left and walks 6 km. Again he turns right and walks 9 km. Finally, he turns to his right and walks 6 km. How far is he from the starting point?
A) 26 km B) 21 km C) 14 km D) 9 km E) None of these

Q15. Sundar runs 20 m towards East and turns to right and runs 10 m. Then he turns to the right and runs 9 m. Again he turns to right and runs 5 m. After this he turns to left and runs 12 m and finally he turns to right and 6 m. Now to which direction is Sundar facing?
A) East B) West C) North D) South E) None of these

Q16. Sachin walks 20 km towards North. He turns left and walks 40 km. He again turns left and walks 20 km. Finally, he moves 20 km after turning to the left. How far is he from his starting position?
A) 20 km. B) 30 km. C) 50 km. D) 60 km. E) None of these

Q17. From his house, Lokesh went 15 km to the North. Then he turned west and covered 10 km. Then he turned south and covered 5 km. Finally turning to the east, he covered 10 km. In which direction is he from his house?
A) East B) West C) North D) South E) None of these

Q18. Village Q is to the North of the village P. The village R is in the East of Village Q. The village S is to the left of the village P. In which direction is the village S with respect to village R?
A) West B) South-West C) South D) North-West
E) None of these

Q19. A's father's mother-in-law's only daughter's son is B. How is A related to B?
A) Brother B) Sister C) Nephew D) Can't say
E) None of these

Q20. Pointing to a person, Raju said, "He is the only brother of my father's mother's daughter." How is the person related to Raju?
A) Brother B) Father C) Uncle D) Nephew
E) None of these

Q21. A's mother's father is the husband of B's mother. How is A related to B, if A and B are both male?
A) Uncle B) Father C) Nephew D) Son
E) None of these

Q22. Tinku, introducing a person to Rinku, said "He is the father of your sister's son and he is also my mother's husband". How is Tinku's father related to Rinku's mother?

A) Nephew B) Uncle C) Son-in-law D) Father
E) None of these

Directions (Q23-Q27): Use the relations defined below and answer the following questions.

A + B means A is the mother of B.
A − B means A is the sister of B.
A × B means A is the father of B.
A ÷ B means A is the son of B.
A = B means A is the brother of B.
A ≠ B means A is the daughter of B.

Q23. Which of the following means P is the aunt of Q?

A) P − R ÷ Q B) P + R × Q C) P ≠ R × Q
D) P − R + Q E) None of these

Q24. Which of the following means, S is the son of T's daughter?

A) T × M + S + N B) T × M + S = N C) T + M × S − N
D) S ÷ M ÷ T − N E) None of these

Q25. Which of the following means W is the uncle of Z?

A) W × A − B + Z B) W = A + B − Z C) W = A + B + Z
D) W − A + B − Z E) None of these

Q26. Which of the following means C is the grandfather of both D and E?

A) C × A ÷ D − E B) C + A + D ≠ E C) C ÷ A ≠ D = E
D) C × A × D − E E) C ÷ A × D × E

Q27. Which of the following means I is the mother of L?

A) I + B − C ≠ D × L B) I ≠ B + C × L C) I + B × C ≠ D − L
D) I + B − C × L E) I × B × C = D = L

Direction (Q28-Q32): In a family of three generations, there are eight members, M and N are brothers. R is the grandson of Q. T is the only niece of M, P is the sister-in-law of N. O is the mother-in-law of S. There are three married couples in the family. T is the daughter of S.

Q28. How is R related to T?
A) Cousin B) Uncle C) Brother D) Father
E) Can't be determined

Q29. How is Q related to S?
A) Grandfather B) Uncle C) Father-in-law D) Father
E) None of these

Q30. How is P related to O?
A) Aunt B) Mother C) Daughter-in-law D) Mother-in-law
E) None of these

Q31. How is O related to R?
A) Uncle B) Grandmother C) Aunt D) Grandfather
E) None of these

Q32. In the family the ratio of number of male members to that of female members
(A) 3 : 5 (B) 5 : 3 (C) 1 : 1 (D) 2 : 3 (E) None of these

Direction (Q33-Q36): A, B, C, D, E and F are six members of a family. A is the mother of B, who is the husband of D. F is the brother of one of the parents of C. D is the daughter-in-law of E and has no siblings (Real Brother and Sister). C is the son of D.

Q33. How is C related to A?
A) Nephew B) Son-in-law C) Grandson D) Father
E) None of the above

Q34. How is F related to D?
A) Cousin B) Brother-in-law C) Brother D) Father
E) None of these

Q35. How is E related to F?
A) Mother B) Mother-in-law C) Father D) Father-in-law
E) None of these

Q36. How many male members are there in the family?
A) Two B) Three C) Four D) Five E) Cannot be determined

Q37. Which year will have the same calendar as that of 2005?
A) 2006 B) 2007 C) 2008 D) 2011 E) 2012

Q38. If you were born on 14th April, 1992, which was a Sunday, then on which day of the week does your birthday fall in 1993?
A) Monday B) Tuesday C) Wednesday D) Friday
E) None of these

Q39. Which year will have same calendar as 1984?
A) 2020 B) 2008 C) 2012 D) 2004
E) None of these

Q40. In a leap year, which month will have the same calendar as that of January in that year?
A) April B) July C) October D) March
E) None of these

Q41. In a year, if 23rd November is a Friday then 14th March in that year is on which day of the week?
A) Monday B) Wednesday C) Sunday D) Tuesday
E) None of these

Q42. Find the angle between the minute hand and the hours hand at 10:40?
A) 90 B) 85 C) 80 D) 75 E) None of these

Q43. What is the angle between the hour and minute hands of a clock 30 minutes after they meet for the third time in a day?
A) 150 B) 165 C) 180 D) None of these
E) Cannot be determined

Q44. A clock gains 15 minutes per day. If it is set right at 12 noon, the time it shows at 4 AM is
A) 4:20 B) 4:30 C) 4:02 D) 4:10 E) None of these

Q45. A clock is set at 5 am. If the clock loses 16 minutes in 24 hours, what will be the true time when the clock indicates 10 pm on 4th day?
A) 9:30 pm B) 10 pm C) 10:30 pm D) 11 pm
E) None of these

Q46) From 1 O' clock afternoon up to 10 O' clock in the night, the hands of a clock be together?
A) 10 B) 9 C) 18 D) 20 E) None of these

Q47. A student has to obtain 45% of the total marks to pass. He got 105 marks and failed by 30 marks. The maximum marks are:
A) 300 B) 500 C) 800 D) 1000 E) None of These

Q48. Two successive increases of 20% and 40% are equal to
A) 49% B) 70% C) 68% D) 52% E) None of These

Q49. If selling price is doubled, the profit becomes 5 times. Find the profit percent?
A) 25% B) 30% C) 33.33% D) 40% E) None of These

Q50. If 12% of x is equal to 6% of y, then 18% of x will be equal to how much % of y?
A) 7% B) 9% C) 11% D) 12% E) None of these

Q51. If a number is 20% more than the another, how much % is the smaller number less than the first?
A) 12.5% B) 16.66% C) 11.11% D) 8.33% E) None of these

Q52. A man's wages were decreased by 50%. Again, the reduced wages were increased by 50%. He has a loss of?
A) 35% B) 25% C) 20% D) 15% E) None of these

Q53. The price of cooking oil has increased by 25%. The percentage of reduction that a family should effect in the use of cooking oil so as not to increase the expenditure on this account is?
A) 35% B) 25% C) 20% D) 15% E) None of these

Q54. A shopkeeper marks the price of his goods at 25% higher than the original price. After that, he allows a discount of 12%. What profit or loss does he get?
A) 10% profit B) 15% profit C) 10% loss D) 15% loss
E) None of these

Q55. A sweet seller sells 3/5th part of sweets at a profit of 10% and remaining at a loss of 5%. If the total profit is Rs 1500, then what is the total cost price of sweets?
A) Rs 36,500 B) Rs 37,000 C) Rs 37,500 D) None of these

Q56. Two articles were sold at same price. On one article there is profit of 20% while on the other there is loss of 20%. Find the net profit or loss %.
A) No profit/ No loss B) 4% profit C) 4% loss D) 5% loss
E) None of these

Q57. How many different values can x take if the number 2506x8 is divisible by 8?
A)1 B) 2 C) 3 D) 4 E) None of these

Q58. If the number 42573x is divisible by 72, find the value that x can assume?
A)6 B) 2 C) 3 D) 4 E) None of these

Q59. What is the Smallest Number that should be added to 89357 to make it divisible by '9'?
A)1 B) 7 C) 3 D) 4 E) None of these

Q60. What is the value of M and N respectively? If M39048458N is divisible by 8 and 11; where M and N are single digit integers?
A) 7, 8 B) 8, 6 C) 6, 4 D) 5, 4 E) None of these

Q61. The difference in simple interest and compound interest on a certain sum of money in 2 years at 12.5 % p.a. is Rs. 50. The sum is
A) Rs. 1000 B) Rs. 6400 C) Rs. 3200 D) Rs. 2000
E) None of these

Q62. The compound interest on a certain sum of money for 2 years is Rs. 208 and the simple interest for the same time at the same rate is Rs. 200. Find the rate %.
A) 5 % B) 6 % C) 7 % D) 8 % E) None of these

Q63.The simple interest on a certain sum for 3 years in Rs. 225 and the compound interest on the same sum for 2 years is Rs. 165. Find the rate percent per annum.
A) 20 % B) 2.5 % C) 5 % D) 15 % E) None of these

Q64. Find the difference between SI and CI for two years if Rs 2000 is invested at the rate of 5% p.a.c.a.
A) Rs 50 B) Rs 20 C) Rs 10 D) Rs 5 E) None of these

Q65. If a sum of money is invested for 3 years at simple interest. If the interest rates applied are 10%, 20%, 30% for first year, second year and third year respectively. Find the sum of money invested if interest received is equal to 120.
A) Rs 500 B) Rs 200 C) Rs 100 D) Rs 50 E) None of these

Q66. Find the simple interest on Rs 68000 at 16 % per annum for 9 months?
A) Rs 8500 B) Rs 8160 C) Rs 8000 D) Rs 5500
E) None of these

Q67. A sum of Rs 12,500 amounts to Rs 15,500 in 4 years at the rate of simple interest. What is the rate of interest?
A) 5 % B) 6 % C) 7.5 % D) 8 % E) None of these

Q68. A person borrows Rs 5000 for 2 years at 4 % p.a. simple interest. He immediately lends it to another person at 6 % p.a. for 2 years. Find his gain in the transaction per year?
A) Rs 112.5 B) Rs 125 C) Rs 100 D) Rs 167.5 E) None of these

Q69. Nitin borrowed some money at the rate of 6% p.a. for the first three years, 9% p. a. for the next five years and 13% p.a. for the period beyond eight years. If the total interest paid by him at the end of eleven years is Rs 8160, how much money did he borrow?
A) Rs 8000 B) Rs 10,000 C) Rs 12,000 D) Data inadequate
E) None of these

Q70.The population of Jhumri Tilaiya increases by 10% in the first year, it increases by 20% in the second year and due to mass exodus, it decreases by 5% in the third year. What will be its population after 3 years, if today it is 200,000?
A)11540 B)13860 C)12860 D) 12540
E) None of these

Q71. Seth Ankush borrows a sum of Rs 1200 at the beginning of a year. After 4 months, Rs 1800 more is borrowed at a rate of interest double the previous one. At the end of the year, the sum of interest on both the loans is Rs 216. What is the first rate of interest per annum?
A)9% B) 6% C) 8% D) 12% E) None of these

Q72. Mihir's capital is 5/4 times more than Tulsi's capital. Tulsi invested her capital at 50 % per annum for 3 years (compounded annually). At what rate % p.a. simple interest should Mihir invest his capital so that after 3 years, they both have the same amount of capital?
A) 20/3 % B) 10 % C) 50/3 % D) 1.728 %
E) None of these

Q73. The difference in simple interest and compound interest on a certain sum of money in 3 years at 10 % p.a. is Rs. 372. The sum is
A) Rs. 8000 B) Rs.9000 C) Rs. 10000 D) Rs. 12000
E) None of these

Q74. If the compound interest on a sum for 2 years at 25/2% per annum is Rs 510, the simple interest on the same sum at the rate for the same period of time is?
A) Rs 400 B) Rs 450 C) Rs 460 D) Rs 480
E) None of these

Q75. If the simple interest on a sum of money for 2 years at 5% per annum is Rs 50, what is the compound interest on the same sum at the same rate and for the same time?
A) Rs 51.25 B) Rs 52 C) Rs 54.25 D) Rs 60
E) None of these

Q76. If a sum of money is borrowed from a bank at 20% p.a.c.a. and returned back in two equal annual instalments of Rs 3600. Find the sum
A) Rs. 5000 B) Rs. 8500 C) Rs. 5500 D) Rs. 6000
E) None of these

Q77. If Rs 8400 is borrowed from a bank at 10% p.a.c.a. and returned back in two equal annual instalments. Find the value of instalments
A) Rs. 5000 B) Rs. 8500 C) Rs. 4840 D) Rs. 6000
E) None of these

Q78. Find the value of equal monthly instalments to be paid at 60% per annum if the amount due in four months is Rs 4300.
A) Rs. 1000 B) Rs. 1200 C) Rs. 1500 D) Rs. 600
E) None of these

Q79. Find the Rate of interest per month if amount due is Rs.2400 after 3 months on principle of Rs.1800.
A) 10% B) 9.09% C) 11.11% D) 5% E) None of these

Q80. In how many years will a sum of Rs. 800 at 10% per annum compounded semi-annually become Rs. 926.10?
A) 4/3 B) 3/2 C)7/2 D)5/2 E) None of these

Printed Pages:04 **University Roll No.....**

Mid-Term Examination, Even Semester 2020-21

B. Tech (Data Analytics), Year-II, Semester-IV

BCSE0552: Applied Statistical Analysis

Time: 2 Hours **Maximum Marks: 30**

Section- A

Note: Attempt All Three Questions. 3 x 2 = 6 Marks

Three questions of 2 marks each.

I. What do you infer from the statements P(A)=0 and P(A)=1? Define mutually exclusive events with an example.

II. Indicate which level of measurement is being used in the given scenario

1. The teacher of a class of third graders records the height of each student.

2. The teacher of a class of third graders records the eye color of each student.

3. The teacher of a class of third graders records the letter grade for mathematics for each student.

4.The roster of a basketball team lists the jersey numbers for each of the players.

III. 1. Identify the Population: Gwinnett County Public Schools randomly selected 230 teachers to find out which technology resource is the most effective. 30 teachers chose Safari Montage, 45 selected Learn Zillion, 100 chose Ed Puzzle, and 55 chose Kahoot. GCPS concluded that all teachers prefer Ed Puzzle.

2. Identify the population: Members of the Drama Club plan to sell popcorn as a fundraiser for their Shakespeare production. They survey 75 students at random about their favorite flavors of popcorn. 33 students like butter, 15 like cheese, and 27 like caramel.

3.Identify the Sample: A restaurant wants to know if customers buy dessert when they eat out. As people leave the restaurant one evening, 20 people are surveyed at random. Eight people say they usually order dessert when they eat out. The restaurant concluded that most customers do not order dessert.

4. Identify the Sample: To determine if the class understood the homework assignment, the math teacher checks the top 3 papers in the pile of collected homework. The teacher finds that all students understood the homework assignment.

Section- B

Note: Attempt All Three Questions. 3 x 3 = 9 Marks

Three questions of 3 marks each .

I. 1. The mean of the following distribution is 26. Find the value of p and also the value of the observation.

xi	0	1	2	3	4	5
fi	3	3	p	7	p-1	4

Also, find the mode and the given data.

2. The weights in kg of 10 students are given below:

39, 43, 36, 38, 46, 51, 33, 44, 44, 43

Find the mode of this data. Is there more than 1 mode? If yes, why?

II. Here are the ages of 50 members of a country social service program:

83 51 66 61 82 65 54 56 92 60
65 87 68 64 51 70 75 66 74 68
44 55 78 69 98 67 82 77 79 62
38 88 76 99 84 47 60 42 66 74
91 71 83 80 68 65 51 56 73 55

Use these data to construct relative frequency distributions using 7 equal intervals and 13 equal intervals. State policies on social service programs require that approximately 50 percent of the program participants be older than 50.

(a) Is the program in compliance with the policy?

(b) Does your 13-interval relative frequency distribution help you answer part (a) better than your 7-interval distribution?

III. The probability that A hits a target is 1/4, and the probability that B hits the target is 2/5. Both shoot at the target. Find the probability that at least one of them hits the target, i.e., that A or B (or both) hit the target.

Section – C

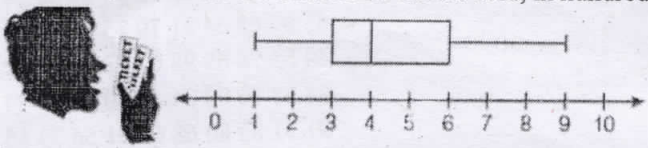
Note: Attempt Any Three Questions. 3 x 5 = 15 Marks

Four questions of 5 marks each.

I. Of the students in the college, 60% of the students reside in the hostel and 40% of the students are day scholars. Previous year result reports that 30% of all students who stay in the hostel scored A Grade and

20% of day scholars scored A grade. At the end of the year, one student is chosen at random and found that he/she has an A grade. What is the probability that the student is a hostler?

II. A movie theater recorded the number of tickets sold daily for a popular vampire movie during the month of December. The box plot below shows the data for the number of tickets sold, in hundreds.



Determine whether the following statements are TRUE or FALSE regarding this box plot. (with valid reason)

1. 25% of the data is between 300 and 400
2. the second quartile is 600
3. the median is 400
4. the first quartile is 300
5. the range is 300
6. 50% of the data is between 300 and 600

III. On an average a typist makes 2 mistakes per page. What is the prob. that she will make (i) No errors on a page (ii) 4 or more errors on a particular page?

IV. Out of 800 families with 4 children each, how many families would be expected to have (i) 2 boys and 2 girls (ii) Children of both genders. Assume equal prob. for boys and girls.

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Mid-Term Examination, Even Semester 2020-21

Programme: B-Tech **Branch:** CCV **Year:** II

Introduction to Virtualization & Cloud Computing, BCSC 0502

Time : 2 Hours **Maximum Marks: 30**

Section- A

Note: Attempt All Three Questions. 3 x 2 = 6 Marks

1. What are the characteristics of cloud computing infrastructure? Briefly explain each characteristic with an example.
2. Write down the four major parameters to compare between cloud and traditional computing.
3. Cloud computing reduces the CAPEX and OPEX of an organization. Explain with key points.

Section- B

Note: Attempt All Three Questions. 3 x 3 =9 Marks

1. Briefly explain the three key computing technologies which became the reason behind evolution of cloud computing namely- cluster computing, grid computing and utility computing.
2. Differentiate between full virtualization and para-virtualization with examples.
3. Discuss different types of storage virtualization.

Section – C

Note: Attempt Any Three Questions. 3 x 5 = 15 Marks

1. What are the shortcomings of traditional IT infrastructure? Also, discuss how Virtualization and cloud helped to remove shortcomings of traditional IT infrastructure?
2. Identify the types of virtualization an organization must use to set up its offices in different part of the world. Justify your answer with proper reasoning. Also, discuss the cost impacts of proposed virtualization.
3. It is challenging to equip an IT resource to match its processing requirements. If the demand for the IT resource is below its capacity, then it is under-utilized and if the demand is above its capacity it is over-utilized or unable to meet the demand. Develop a service model to show how can IT resources be scaled automatically in response to fluctuating demand?
4. Explain the features of virtual desktop infrastructure. Why we need VDI?

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Mid-Term Examination, Even Semester 2020-21

Programme: B-Tech **Branch:** IoT **Year:** II

Cloud Architecture & Deployment Models, BCSE 0652

Time: 2 Hours **Maximum Marks: 30**

Section- A

Note: Attempt All Three Questions. **3 x 2 = 6 Marks**

I. Any request you made to a server over internet is transferred in plain text form. Suggest a networking technique that provides more security and also help in hide your identity. Compare the suggested technique with network proxy.

II. Compare desktop virtualization with the terminal services.

III. What technique is used to reduce the broadcast domain of a network? Explain with the help of example.

Section- B

Note: Attempt All Three Questions. **3 x 3 = 9 Marks**

I. Explain the differences between cloud & virtualization. Also, explain the fields where cloud and virtualization overlap.

II. A cloud solution composed of several major components. Explain all such components in details.

III. Define Cloud Bursting? Also, what are the different solutions provided by hybrid cloud to avoid the problem of vendor lock-in?

Section – C

Note: Attempt Any Three Questions. **3 x 5 = 15 Marks**

I. Explain various types of RAID levels. Also, compare these RAID levels based on their advantages and disadvantages.

II. Compare full virtualization with para virtualization? Also, compare hardware assisted virtualization with software assisted virtualization.

III. Explain XaaS. Also, explain at least five different types of services that fall in the category of XaaS?

IV. Explain with the help of example, how cloud computing is related to client server model, cluster computing, grid computing and utility computing?

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Mid-Term Examination, Even Semester 2020-21

B.Tech (CSE-CSF), II Year, IV Semester

BCSE 0602: IT Network Security

Time : 2 Hours **Maximum Marks: 30**

Section- A

Note: Attempt All Three Questions. 3 x 2 = 6 Marks

Q1. Differentiate

a. Hub and Switch

b. Router and Gateway

Q2. Differentiate

a. Horizontal and Vertical privilege escalation

b. DHCP spoofing and DHCP starvation

Q3. A company is granted the site address 156.96.0.0. The company needs 1700 subnets. Show the complete process of designing the subnets and calculate the subnet mask.

Section- B

Note: Attempt All Three Questions. 3X3=9 Marks

Q1. Discuss the well-known mechanism for distributing the public key globally to users. Who will generate the required keys for achieving the digital signature.

Q2. What do you mean by hash algorithm? To achieve which basic network security aspect, hash algorithms are used. Calculate the

number of padding bits required for SHA-512 algorithm having message length of 3000 bits.

Q3. What do you mean by physical security? Consider a situation where you were given the task of designing physical security for any software organization. Give the four most important points that you will consider for providing the physical security in such a scenario.

Section – C

Note: Attempt Any Three Questions. 3 x 5 = 15 Marks

Q1. What is the basic difference between the SSL and TLS? Give the various protocols used in SSL. Discuss the SSL record protocol in detail.

Q2. Differentiate authentication and authorization. Explain the different types of authentication in detail. What is the importance of compression and email compatibility for encrypting the e-mail?

Q3. What is host security? Discuss the three basic security baselines for achieving host security. Explain how web server and router can be hardened?

Q4. What is the use of network security policy? Discuss its various advantages in detail. Design the End User's account and User's password policy for any organization.

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Mid-Term Examination, Even Semester 2020-21

B.Tech. (CSF, CCV, DA, IoT, AIML), 2nd Year, 4th Semester

Applied Database Management System-BCSC 0014

Time :2 Hours **Maximum Marks: 30**

Section- A

Note: Attempt All Three Questions. 3 x 2 = 6 Marks

1. Consider a relation R(A,B,C,D,E)with the following dependencies: AB->C,CD->E,DE->B.Is AB a candidate key of this relation? if not is ABD?
2. What is entity type, entity set? Explain the difference among an entity an entity type and an entity set.
3. Differentiate between single valued attribute vs multiple valued attributes with suitable diagram.

Section- B

Note: Attempt All Three Questions. 3 x 3 =9 Marks

1. Discuss insertion deletion and modification anomalies? Illustrate with example.
2. Consider the following schema, and write the following query in Relational Algebra
   Employee (emp_id, emp_name, emp_street, emp_city)
   Works (emp_id, company_id, salary)
   Located_in (company_code, company_name, company_city)
   i. Find the names of the emp who work in company 'SANOMED'.
   ii. Find the name of all employees who work for a company located at 'LUCKNOW'.
   iii. Find the names of the employees who do not work for company 'SANOMED'.

iv. Find the names of the employees who work for the company located at city in which they live.

v. Delete the record of emp id 1001.

vi. Add <1006,'jay','xyz','delhi'> in emp table

3. What do you understand by data independence and also explain the types of data independence?

Section – C

Note: Attempt Any Three Questions. 3 x 5 = 15 Marks

1. Consider the following two relation R and S.

R

P	Q	R
10	a	5
15	b	8
25	a	6

S

A	B	C
10	b	6
25	c	3
10	b	5

Then find the following

i) $R \bowtie_{R.P=S.A} S$ ii) $R \bowtie_{R.Q=S.B} S$ iii) $R \rtimes_{R.P=S.A} S$ iv) $R \cup S$ v) $R \cap S$

2.

i. What is the role database administrator? Explain each role in brief.

ii. Explain the first, second and third normal form with example?

3. In a relation schema R(ABCDEH), if F = {A→BCD, CD→E, E→CD, D→AH, ABH→BD, DH→BC} then find out canonical Cover of F.

4.

i. Given R(ABCDE) with the functional dependencies R(AB→CD,A→E,C→D) then check the decomposition R1(ABC),R2(BCD),R3(CDE) is loss less or not.

ii. What do you understand by DDL and DML?

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Mid-Term Examination, Even Semester 2020-21

B. Tech. CSE(AIML), II Year, IV Semester

BCSE 0702: Algorithm for Intelligent Systems & Robotics

Time: 2 Hours **Maximum Marks: 30**

Section- A

Note: Attempt All Three Questions. 3 x 2 = 6 Marks

1. Give a PEAS description of Automated Taxi Driving System.
2. What do you understand by Admissible Heuristic?
3. Explain the dimensions along which the search strategies are evaluated.

Section- B

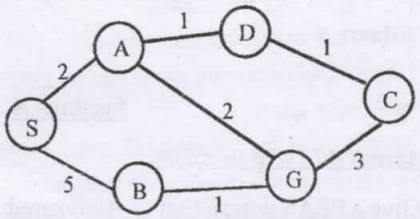
Note: Attempt All Three Questions. 3 x 3 = 09 Marks

1. Consider a state space where start state is number 1 and successor function for state n returns two states, number 2n and 2n+1.
   a) Draw the portion of the state space for states 1 to 15. [1 marks]
   b) Suppose the goal state is 11. List the order in which nodes will be visited using Depth First Search. [2 marks]
2. When does Hill Climbing fail? Also explain the measures to overcome them.
3. Using the inference rules show that **'S'** can be inferred if the following sentences are believed.
   i. $P \rightarrow \neg Q$
   ii. $Q \vee R$
   iii. $\neg S \rightarrow P$
   iv. $\neg R$

Section – C

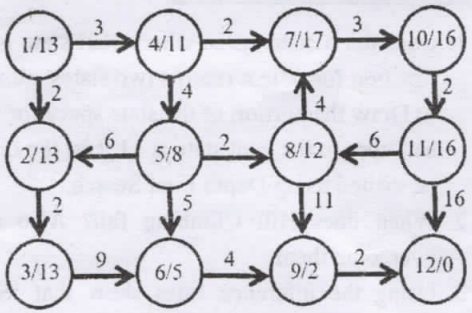
Note: Attempt Any Three Questions. 3 x 5 = 15 Marks

1. Given the following graph where S is the starting node and G is the goal node. Numbers related to arcs between nodes indicate step costs.



Trace graph-search algorithm using BFS strategy. Show the order in which nodes are added to the fringe as well as the generated tree and the solution path. Indicate if found solution is optimal.

2. Describe the Hill Climbing Search Technique.
3. Apply A* Search to the following graph to find minimum cost path from node 1 to node 12.



4. When an inference rule is said to be sound and complete? Find out whether the propositional formula $[(P \rightarrow (Q \vee R)) \rightarrow ((P \wedge Q) \rightarrow R)]$ is valid, satisfiable or contradiction. [2+3=5 marks]